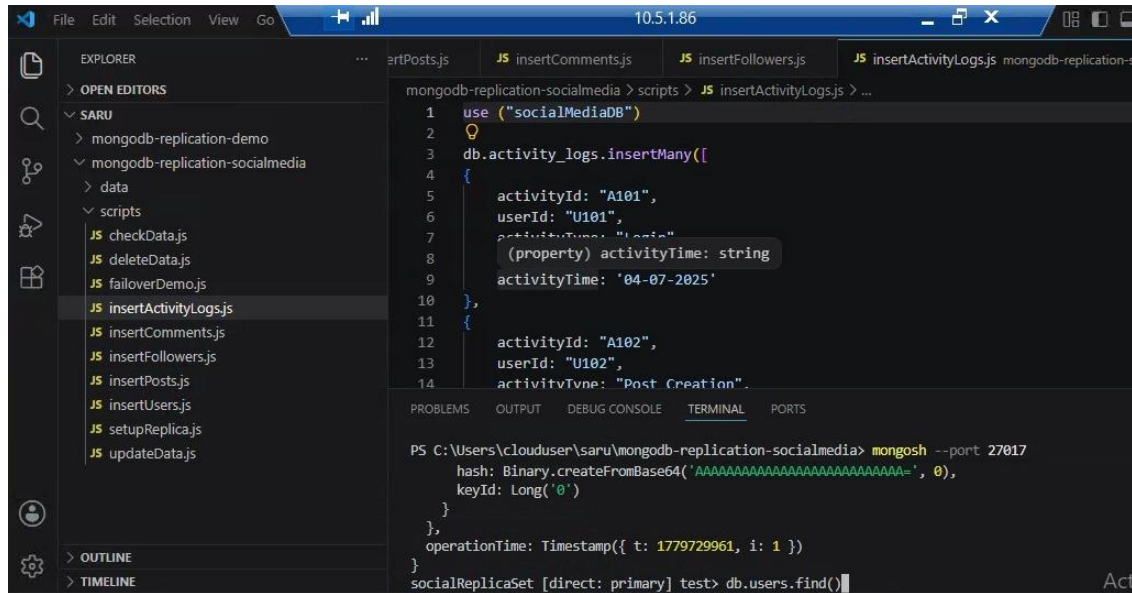


Social Media Platform Case Study Assignment – MongoDB Replication & Replica Sets

Folder Structure:



The screenshot shows the Visual Studio Code interface. The Explorer pane on the left displays the project structure:

- EXPLORED
- OPEN EDITORS
- SARU
 - mongodb-replication-demo
 - mongodb-replication-socialmedia
 - data
 - scripts
 - checkData.js
 - deleteData.js
 - failoverDemo.js
 - insertActivityLogs.js
 - insertComments.js
 - insertFollowers.js
 - insertPosts.js
 - insertUsers.js
 - setupReplica.js
 - updateData.js

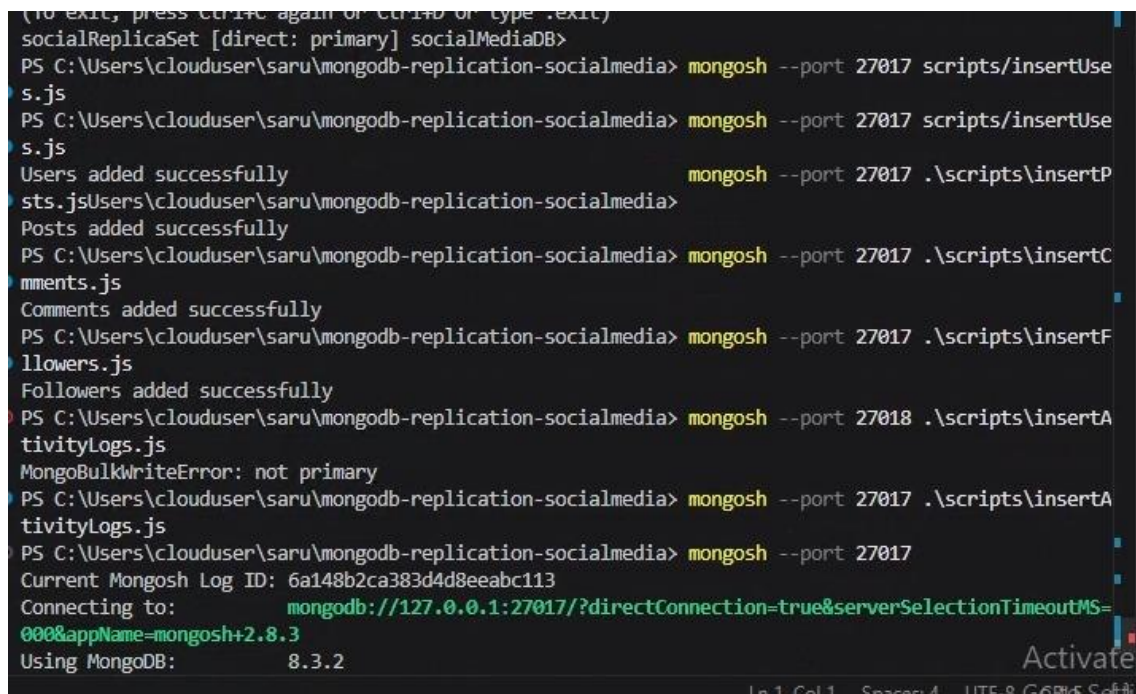
The main editor shows the content of `insertActivityLogs.js`:

```
1 use ("socialMediaDB")
2
3 db.activity_logs.insertMany([
4   {
5     activityId: "A101",
6     userId: "U101",
7     activityType: "login",
8     (property) activityTime: string
9     activityTime: '04-07-2025'
10  },
11  {
12    activityId: "A102",
13    userId: "U102",
14    activityType: "Post Creation".
```

The bottom pane shows the TERMINAL output:

```
PS C:\Users\clouduser\saru\mongodb-replication-socialmedia> mongosh --port 27017
hash: Binary.createFromBase64('AAAAAAAAAAAAAAAAAAAAAAAAAAAA', 0),
keyId: Long('0')
},
operationTime: Timestamp({ t: 1779729961, i: 1 })
}
socialReplicaSet [direct: primary] test> db.users.find()
```

Create Operations:



```
(to exit, press Ctrl+C again or Ctrl+D or type .exit)
socialReplicaSet [direct: primary] socialMediaDB>
PS C:\Users\clouduser\saru\mongodb-replication-socialmedia> mongosh --port 27017 scripts/insertUse
s.js
PS C:\Users\clouduser\saru\mongodb-replication-socialmedia> mongosh --port 27017 scripts/insertUse
s.js
Users added successfully
PS C:\Users\clouduser\saru\mongodb-replication-socialmedia> mongosh --port 27017 .\scripts\insertP
sts.js
Posts added successfully
PS C:\Users\clouduser\saru\mongodb-replication-socialmedia> mongosh --port 27017 .\scripts\insertC
mments.js
Comments added successfully
PS C:\Users\clouduser\saru\mongodb-replication-socialmedia> mongosh --port 27017 .\scripts\insertF
llowers.js
Followers added successfully
PS C:\Users\clouduser\saru\mongodb-replication-socialmedia> mongosh --port 27018 .\scripts\insertA
tivityLogs.js
MongoBulkWriteError: not primary
PS C:\Users\clouduser\saru\mongodb-replication-socialmedia> mongosh --port 27017 .\scripts\insertA
tivityLogs.js
PS C:\Users\clouduser\saru\mongodb-replication-socialmedia> mongosh --port 27017
Current Mongosh Log ID: 6a148b2ca383d4d8eeabc113
Connecting to: mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=
000&appName=mongosh+2.8.3
Using MongoDB: 8.3.2
```

Read Operations:

1. View Users:

```
PS C:\Users\clouduser\saru\mongodb-replication-socialmedia> mongosh --port 27017
socialReplicaSet [direct: primary] socialMediaDB> db.users.find()
[
  {
    _id: ObjectId('6a1489e2650c35be86abc114'),
    userId: 'U101',
    username: 'saru',
    email: 'saru@gmail.com',
    bio: 'Frontend Developer',
    followersCount: 120,
    followingCount: 80,
    createdAt: '25-11-2015'
  },
  {
    _id: ObjectId('6a1489e2650c35be86abc115'),
    userId: 'U102',
    username: 'kavi',
    email: 'kavi@gmail.com',
    bio: 'Travel Blogger',
    followersCount: 300,
    followingCount: 150,
    createdAt: '08-08-2010'
  },
  {
    _id: ObjectId('6a1489e2650c35be86abc116'),
    userId: 'U103',
    username: 'priya',
    email: 'priya@gmail.com',
    bio: 'Software Engineer',
    followersCount: 50,
    followingCount: 20,
    createdAt: '12-03-2018'
  }
]
```

2. View Posts:

```
socialReplicaSet [direct: primary] socialMediaDB> db.posts.find()
[
  {
    _id: ObjectId('6a148ab64868747248abc114'),
    postId: 'P101',
    userId: 'U101',
    caption: 'Learn Python',
    hashtags: [ '#python', '#functions' ],
    likesCount: 250,
    createdAt: '15-01-2024'
  },
  {
    _id: ObjectId('6a148ab64868747248abc115'),
    postId: 'P102',
    userId: 'U102',
    caption: 'Beautiful beach vibes',
    hashtags: [ '#travel', '#beach' ],
    likesCount: 450,
    createdAt: '01-01-2020'
  },
  {
    _id: ObjectId('6a148ab64868747248abc116'),
    postId: 'P103',
    userId: 'U103',
    caption: 'A day in the life of a developer',
    hashtags: [ '#coding', '#life' ],
    likesCount: 100,
    createdAt: '20-05-2023'
  }
]
```

3. View Comments:

```
]
socialReplicaSet [direct: primary] socialMediaDB> db.comments.find()
[
  {
    _id: ObjectId('6a148ac2242f0071edabc114'),
    commentId: 'C101',
    postId: 'P101',
    userId: 'U102',
    commentText: 'Great learning!',
    commentedAt: '02-02-2025'
  },
  {
    _id: ObjectId('6a148ac2242f0071edabc115'),
    commentId: 'C102',
    postId: 'P102',
    userId: 'U101',
    commentText: 'Amazing view',
    commentedAt: '02-01-2025'
  }
]
```

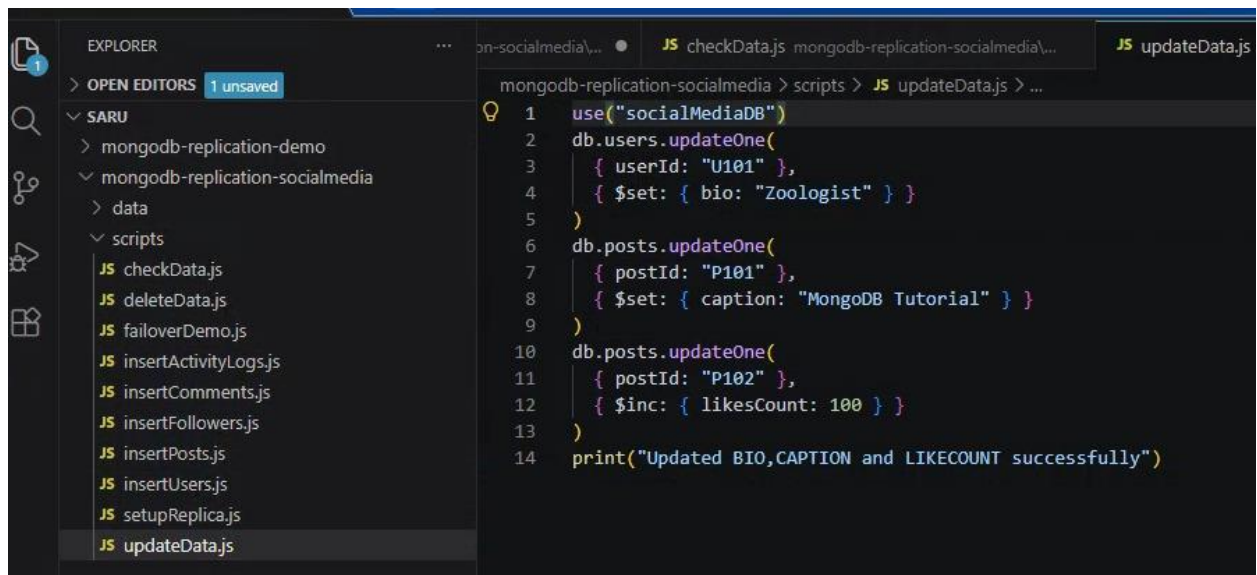
4. View Followers:

```
]
socialReplicaSet [direct: primary] socialMediaDB> db.followers.find()
[
  {
    _id: ObjectId('6a148acebb82a4afecabc114'),
    followId: 'F101',
    followerId: 'U101',
    followingId: 'U102',
    followedAt: '01-01-2026'
  },
  {
    _id: ObjectId('6a148acebb82a4afecabc115'),
    followId: 'F102',
    followerId: 'U102',
    followingId: 'U101',
    followedAt: '02-01-2026'
  }
]
```

5. View ActivityLogs:

```
socialReplicaSet [direct: primary] socialMediaDB> db.activity_logs.find()
[
  {
    _id: ObjectId('6a148af6c69d8750deabc114'),
    activityId: 'A101',
    userId: 'U101',
    activityType: 'Login',
    device: 'Mobile',
    activityTime: '04-07-2025'
  },
  {
    _id: ObjectId('6a148af6c69d8750deabc115'),
    activityId: 'A102',
    userId: 'U102',
    activityType: 'Post Creation',
    device: 'Laptop',
    activityTime: '01-01-2023'
  }
]
```

Update Operations:



The screenshot shows a Visual Studio Code editor with a file explorer on the left and a code editor on the right. The file explorer shows a project structure with folders 'SARU', 'mongodb-replication-demo', 'mongodb-replication-socialmedia', 'data', and 'scripts'. The 'scripts' folder is expanded, showing several JavaScript files including 'updateData.js'. The code editor displays the content of 'updateData.js', which contains MongoDB update operations for users and posts.

```
1 use("socialMediaDB")
2 db.users.updateOne(
3   { userId: "U101" },
4   { $set: { bio: "Zoologist" } }
5 )
6 db.posts.updateOne(
7   { postId: "P101" },
8   { $set: { caption: "MongoDB Tutorial" } }
9 )
10 db.posts.updateOne(
11   { postId: "P102" },
12   { $inc: { likesCount: 100 } }
13 )
14 print("Updated BIO,CAPTION and LIKECOUNT successfully")
```

```
socialReplicaSet [direct: primary] socialMediaDB>
PS C:\Users\clouduser\saru\mongodb-replication-socialmedia> mongosh --port 27017 .\scripts\updateData.js
Updated BIO,CAPTION and LIKECOUNT successfully
PS C:\Users\clouduser\saru\mongodb-replication-socialmedia> mongosh --port 27017
```

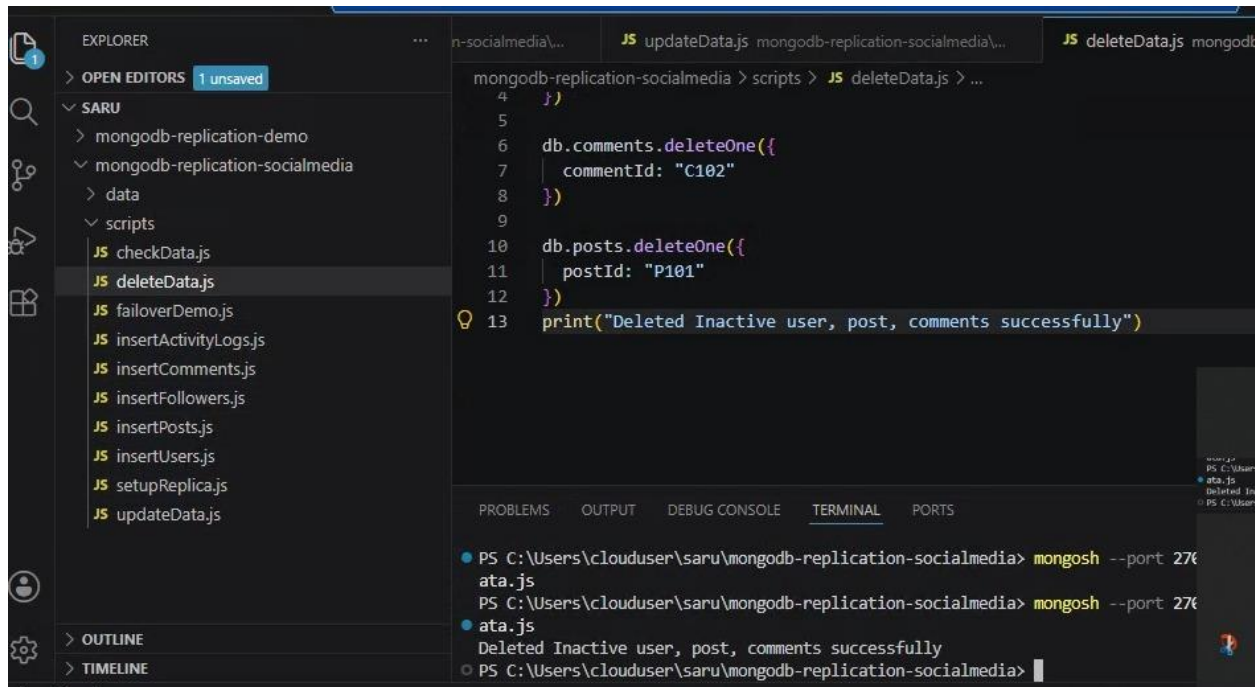

1. Update Bio:

```
PS C:\Users\clouduser\saru\mongodb-replication-socialmedia> mongosh --port 27017
switched to db socialMediaDB
socialReplicaSet [direct: primary] socialMediaDB> use socialMediaDB
already on db socialMediaDB
socialReplicaSet [direct: primary] socialMediaDB> db.users.find()
[
  {
    _id: ObjectId('6a1489e2650c35be86abc114'),
    userId: 'U101',
    username: 'saru',
    email: 'saru@gmail.com',
    bio: 'Zoologist',
    followersCount: 120,
    followingCount: 80,
    createdAt: '25-11-2015'
  },
  {
    _id: ObjectId('6a1489e2650c35be86abc115'),
    userId: 'U102',
    username: 'kavi',
    email: 'kavi@gmail.com',
    bio: 'Travel Blogger',
    followersCount: 300,
    followingCount: 150,
    createdAt: '08-08-2010'
  }
]
```

2. Update Caption & Update Like_Count:

```
]
socialReplicaSet [direct: primary] socialMediaDB> db.posts.find()
[
  {
    _id: ObjectId('6a148ab64868747248abc114'),
    postId: 'P101',
    userId: 'U101',
    caption: 'MongoDB Tutorial',
    hashtags: [ '#python', '#functions' ],
    likesCount: 250,
    createdAt: '15-01-2024'
  },
  {
    _id: ObjectId('6a148ab64868747248abc115'),
    postId: 'P102',
    userId: 'U102',
    caption: 'Beautiful beach vibes',
    hashtags: [ '#travel', '#beach' ],
    likesCount: 550,
    createdAt: '01-01-2020'
  }
]
```

Delete Operations:



The screenshot shows the Visual Studio Code interface. On the left, the Explorer pane shows a project structure with a 'scripts' folder containing several JavaScript files, including 'deleteData.js'. The main editor displays the content of 'deleteData.js', which contains two MongoDB delete operations: one for a comment and one for a post. The terminal at the bottom shows the command prompt where 'mongosh' is used to execute the script, resulting in a confirmation message.

```
mongodb-replication-socialmedia > scripts > JS deleteData.js > ...
4   })
5
6   db.comments.deleteOne({
7     commentId: "C102"
8   })
9
10  db.posts.deleteOne({
11    postId: "P101"
12  })
13  print("Deleted Inactive user, post, comments successfully")
```

Terminal Output:

```
PS C:\Users\clouduser\saru\mongodb-replication-socialmedia> mongosh --port 27018
ata.js
PS C:\Users\clouduser\saru\mongodb-replication-socialmedia> mongosh --port 27018
ata.js
Deleted Inactive user, post, comments successfully
PS C:\Users\clouduser\saru\mongodb-replication-socialmedia>
```

1. Delete inactive users:

```
socialReplicaSet [direct: primary] test> use socialMediaDB
switched to db socialMediaDB
socialReplicaSet [direct: primary] socialMediaDB> db.users.find()
[
  {
    _id: ObjectId('6a1489e2650c35be86abc114'),
    userId: 'U101',
    username: 'saru',
    email: 'saru@gmail.com',
    bio: 'Zoologist',
    followersCount: 120,
    followingCount: 80,
    createdAt: '25-11-2015'
  }
]
```

2. Delete unwanted comments & Delete posts:

```
] socialReplicaSet [direct: primary] socialMediaDB> db.posts.find()
[
  {
    _id: ObjectId('6a148ab64868747248abc115'),
    postId: 'P102',
    userId: 'U102',
    caption: 'Beautiful beach vibes',
    hashtags: [ '#travel', '#beach' ],
    likesCount: 550,
    createdAt: '01-01-2020'
  }
]
socialReplicaSet [direct: primary] socialMediaDB> db.comments.find()
[
  {
    _id: ObjectId('6a148ac2242f0071edabc114'),
    commentId: 'C101',
    postId: 'P101',
    userId: 'U102',
    commentText: 'Great learning!',
    commentedAt: '02-02-2025'
  }
]
socialReplicaSet [direct: primary] socialMediaDB> 
```

Ln 13, Col 58 Spaces: 4 UTF

Replica Set Operations:

```
mongodb-replication-socialmedia > scripts > JS setupReplica.js >
1  rs.initiate({
2    _id:"socialReplicaSet",
3    members:[
4      {_id:0,host:"localhost:27017"},
5      {_id:1,host:"localhost:27018"},
6      {_id:2,host:"localhost:27019"}
7    ]
8  })
```

Set Primary & Secondary Nodes:

```
PS C:\Users\clouduser\saru\mongodb-replication-socialmedia> mongosh --port 27017
newTermStartDate: ISODate('2026-05-25T17:21:55.384Z'),
wMajorityWriteAvailabilityDate: ISODate('2026-05-25T17:21:55.478Z')
},
members: [
  {
    _id: 0,
    name: 'localhost:27017',
    health: 1,
    state: 1,
    stateStr: 'PRIMARY',
    uptime: 3007,
    optime: { ts: Timestamp({ t: 1779732668, i: 4 }), t: Long('1') },
    optimeDate: ISODate('2026-05-25T18:11:08.000Z'),
    optimeWritten: { ts: Timestamp({ t: 1779732668, i: 4 }), t: Long('1') },
    optimeWrittenDate: ISODate('2026-05-25T18:11:08.000Z'),
    lastAppliedWallTime: ISODate('2026-05-25T18:11:08.398Z'),
    lastDurableWallTime: ISODate('2026-05-25T18:11:08.398Z'),
    lastWrittenWallTime: ISODate('2026-05-25T18:11:08.398Z'),
    syncSourceHost: '',
    syncSourceId: -1,
    infoMessage: '',
    electionTime: Timestamp({ t: 1779729715, i: 1 }),
    electionDate: ISODate('2026-05-25T17:21:55.000Z'),
    configVersion: 5,
    configTerm: 1,
    self: true,
```

```
PS C:\Users\clouduser\saru\mongodb-replication-socialmedia> mongosh --port 27017
  _id: 1,
  name: 'localhost:27018',
  health: 1,
  state: 2,
  stateStr: 'SECONDARY',
  uptime: 2724,
  optime: { ts: Timestamp({ t: 1779732668, i: 4 }), t: Long('1') },
  optimeDurable: { ts: Timestamp({ t: 1779732668, i: 4 }), t: Long('1') },
  optimeWritten: { ts: Timestamp({ t: 1779732668, i: 4 }), t: Long('1') },
  optimeDate: ISODate('2026-05-25T18:11:08.000Z'),
  optimeDurableDate: ISODate('2026-05-25T18:11:08.000Z'),
  optimeWrittenDate: ISODate('2026-05-25T18:11:08.000Z'),
  lastAppliedWallTime: ISODate('2026-05-25T18:11:08.398Z'),
  lastDurableWallTime: ISODate('2026-05-25T18:11:08.398Z'),
  lastWrittenWallTime: ISODate('2026-05-25T18:11:08.398Z'),
  lastHeartbeat: ISODate('2026-05-25T18:11:11.186Z'),
  lastHeartbeatRecv: ISODate('2026-05-25T18:11:11.184Z'),
  pingMs: Long('0'),
  lastHeartbeatMessage: '',
  syncSourceHost: 'localhost:27017',
  syncSourceId: 0,
  infoMessage: '',
  configVersion: 5,
  configTerm: 1
},
{
```



```

PS C:\Users\clouduser\saru\mongodb-replication-socialmedia> mongosh --port 27017
{
  _id: 2,
  name: 'localhost:27019',
  health: 1,
  state: 2,
  stateStr: 'SECONDARY',
  uptime: 2714,
  optime: { ts: Timestamp({ t: 1779732668, i: 4 }), t: Long('1') },
  optimeDurable: { ts: Timestamp({ t: 1779732668, i: 4 }), t: Long('1') },
  optimeWritten: { ts: Timestamp({ t: 1779732668, i: 4 }), t: Long('1') },
  optimeDate: ISODate('2026-05-25T18:11:08.000Z'),
  optimeDurableDate: ISODate('2026-05-25T18:11:08.000Z'),
  optimeWrittenDate: ISODate('2026-05-25T18:11:08.000Z'),
  lastAppliedWallTime: ISODate('2026-05-25T18:11:08.398Z'),
  lastDurableWallTime: ISODate('2026-05-25T18:11:08.398Z'),
  lastWrittenWallTime: ISODate('2026-05-25T18:11:08.398Z'),
  lastHeartbeat: ISODate('2026-05-25T18:11:11.186Z'),
  lastHeartbeatRecv: ISODate('2026-05-25T18:11:11.176Z'),
  pingMs: Long('0'),
  lastHeartbeatMessage: '',
  syncSourceHost: 'localhost:27017',
  syncSourceId: 0,
  infoMessage: '',
  configVersion: 5,
  configTerm: 1
}
],

```

Stop Primary node:

```

PS C:\Users\clouduser\saru\mongodb-replication-socialmedia> mongosh --port 27018
members: [
  {
    _id: 0,
    name: 'localhost:27017',
    health: 0,
    state: 8,
    stateStr: '(not reachable/healthy)',
    uptime: 0,
    optime: { ts: Timestamp({ t: 0, i: 0 }), t: Long('-1') },
    optimeDurable: { ts: Timestamp({ t: 0, i: 0 }), t: Long('-1') },
    optimeWritten: { ts: Timestamp({ t: 0, i: 0 }), t: Long('-1') },
    optimeDate: ISODate('1970-01-01T00:00:00.000Z'),
    optimeDurableDate: ISODate('1970-01-01T00:00:00.000Z'),
    optimeWrittenDate: ISODate('1970-01-01T00:00:00.000Z'),
    lastAppliedWallTime: ISODate('2026-05-25T18:14:10.313Z'),
    lastDurableWallTime: ISODate('2026-05-25T18:14:10.313Z'),
    lastWrittenWallTime: ISODate('2026-05-25T18:14:10.313Z'),
    lastHeartbeat: ISODate('2026-05-25T18:14:47.244Z'),
    lastHeartbeatRecv: ISODate('2026-05-25T18:14:14.961Z'),
    pingMs: Long('0'),
    lastHeartbeatMessage: 'Error connecting to localhost:27017 (127.0.0.1:27017) :: caused by :: e
caused by :: No connection could be made because the target machine actively refused it.',
    syncSourceHost: '',
    syncSourceId: -1,
    infoMessage: '',
    configVersion: 5,

```

Activate

Node 27018 becomes Primary:

```
PS C:\Users\clouduser\saru\mongodb-replication-socialmedia> mongosh --port 27018
```

```
{
  _id: 1,
  name: 'localhost:27018',
  health: 1,
  state: 1,
  stateStr: 'PRIMARY',
  uptime: 3120,
  optime: { ts: Timestamp({ t: 1779732890, i: 1 }), t: Long('2') },
  optimeDate: ISODate('2026-05-25T18:14:50.000Z'),
  optimeWritten: { ts: Timestamp({ t: 1779732890, i: 1 }), t: Long('2') },
  optimeWrittenDate: ISODate('2026-05-25T18:14:50.000Z'),
  lastAppliedWallTime: ISODate('2026-05-25T18:14:50.319Z'),
  lastDurableWallTime: ISODate('2026-05-25T18:14:50.319Z'),
  lastWrittenWallTime: ISODate('2026-05-25T18:14:50.319Z'),
  syncSourceHost: '',
  syncSourceId: -1,
  infoMessage: '',
  electionTime: Timestamp({ t: 1779732839, i: 1 }),
  electionDate: ISODate('2026-05-25T18:13:59.000Z'),
  configVersion: 5,
  configTerm: 2,
  self: true,
  lastHeartbeatMessage: ''
},
{
  _id: 2,
```

Activ